

**AMENDMENT NO. 2 JUNE 2008
TO
IS 1179 : 1967 EQUIPMENT OF EYE AND FACE
PROTECTION DURING WELDING**

(First Revision)

[Page 7, clause 6.2(b)] — Insert the following at the end:

- 'c) Gas cutting goggles to be worn over power corrected spectacles shall have a pair of flip-up type cup with screw cap arrangement to hold the filter lenses and protective plain glass.'**

(Page 7, clause 6.2.1) — Substitute the following for the existing:

'The goggles shall be made out of Bakelite/PVC material with smooth finish and shall conform to the requirements of 5.'

(Page 7, clause 6.2.4) — Add the following before the existing first sentence:

'Eye caps shall have two pairs of clear glass in size 50 mm dia. One pair to be fixed in the frame with a screw ring for protecting the eyes while filter lens are lifted up and another pair in the flip-up frame as protective cover to filter lens.'

(Page 9, clause 6.2.8) — Add the following new clauses:

6.3 Gas cutting goggles to be worn over power corrected spectacles.

6.3.1 Goggles shall be made out of bakelite/PVC material with smooth finish, suitable to wear over the spectacle.

6.3.2 They shall have a pair of slip-up type cut with screw cap arrangement to hold the filter lenses and protective plain glass.

6.3.3 They shall have two pairs of clear plain glass in size 50 mm dia. One pair to be fixed in the frame with a screw ring for protecting the eyes while the filter lenses are lifted up and another pair in the flip-up frame as protective cover to filter lenses.

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(Page 9, clause heading 7, and clauses 7.1.2 to 7.1.3 and 7.1.5) — Substitute the words 'Helmet, Head Shield and Hand Shields' for 'Helmet and Hand Shields' wherever appear.

(Page 9, clause 7.2.1) — Substitute the following for the existing:

'Helmets and Shields shall be mould out Fiber material to the thickness of 1.5 mm with clear finish and shall conform to the requirement of 5.'

(Page 10, clause 7.2.4) — Insert the following new clause 7.2.4 and renumber the existing clause 7.2.4 as 7.2.5 after modification as given below:

'7.2.4 It shall have a provision to allow the head shield drop gradually or step by step and have a stopper to arrest the fall and hit at the chest.

7.2.5 The headgear shall have a ratchet type adjustable headband made up of PVC and provided with soft cellulose sponge at the front side of the headband which must be covered by Rexine cloth. It shall retain the helmet comfortably and firmly in place on the wearer's head but shall permit the helmet body to be tilted back over the head. The headgear shall be adjustable, without the use of tools, so as to permit the setting of the filter in the line of vision of the wearer. It shall have the 4 mm wing Nut with spring attachment or any suitable arrangement for easy positioning.'

(Page 10, clause 7.2.6) — Insert the following new clause 7.2.6 and renumber the existing clauses 7.2.5 and 7.2.6 as 7.2.7 and 7.2.8:

'7.2.6 The window shall have a flip-up door arrangement to lift up the filter lens with the provision of clamp to hold the Filter Lens and Protective plain glass.'

(Page 10, clause 7.2.8) — Insert the following new clauses at the end:

7.3 Welding Head Shield

7.3.1 Welding head shield shall be mould out fibre material to the thickness of 1.5 mm with clear finish.

7.3.2 It shall have the 4 mm wing Nut with spring attachment or any suitable arrangement for easy positioning.

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7.3.3 It shall have a ratchet type adjustable headband made up of PVC and provided with soft cellulose sponge at the front side of the headband which must be covered by Rexine cloth.

7.3.4 It shall have a provision to allow the head shield drop gradually or step by step and have a stopper to arrest the fall and hit at the chest.

7.3.5 The total weight of the head shield shall not exceed 400 g and hand shield shall not exceed 350 g.

7.4 Flip-Up Type Welding Helmet

7.4.1 Flip-up type welding helmet shall be mould out of fibre material to the thickness of 1.5 mm.

7.4.2 It shall have the window with the provision of clamp to hold the protective plain glass.

7.4.3 The window shall have a flip-up door arrangement to lift up the filter lens with the provision of clamp to hold the filter lens and protective plain glass.

7.4.4 It shall have the 4 mm wing Nut with spring attachment or any suitable arrangement for easy positioning.

7.4.5 It shall have an ratchet type adjustable headband made up of PVC and provided with soft cellulose sponge at the front side of the headband which must be covered by rexine cloth.

7.4.6 It shall have a provision to allow the head shield drop gradually or step by step and have a stopper to arrest the fall and hit at the chest.

7.4.7 The total weight of the head shield shall not exceed 400 g.

7.5 Auto-Darkening Welding Helmet

The auto-darkening feature of the helmet facilitates to view the workpiece through lens. The lens reacts to the arc instantly and selects the darkness to the selected shade level automatically.

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The selected shade level corresponds to a lens shade number as selected normally depending upon the welding process used for a particular application. The number just denotes the amount of darkness provided by that particular lens and should be used by operators as a guide to select the one that is most comfortable and yet provides good visibility for the particular application.

Welding arcs emit both IR and UV wavelengths of light. Unprotected from this light, both eyes damage and discomfort can occur. Since high-quality auto-darkening helmets provide UV and IR protection even when the helmet is not activated, eyes are always protected. However, for maximum comfort, a high quality helmet that has a response darkening time of 4/10ths of a millisecond is desired. Less than a millisecond is not perceivable by the human eye and will provide the most comfort.

Two varieties, that is, battery-operated and solar-powered of auto-darkening helmets are available. In most cases, it simply is a choice of personal preference and what is most convenient to the operator. These helmets offer a feature that will automatically turn off the battery after the helmet has been kept idle. Solar-powered also offers the convenience of not having to worry about changing batteries.

If the same arc welding process is used on the same material, a fixed shade is sufficient. But if a variety of materials and welding processes are used for a number of different applications, it is better to select a variable shade, which will adjust to the correct darkness for the particular process.

Besides looking for something that is lightweight, the other important consideration would be to find a helmet that has a sensor bar. With this feature, it will limit the field of response so the other welding arcs nearby won't trigger the helmet. A helmet that has a curved clear spatter shield over the optics will provide the best optical clarity. In addition, a helmet that is fully adjustable is important. Some helmets can only adjust the tilt of the helmet to the chest. A fully adjustable helmet also provides adjustment on how close the helmet is located to the face. This is especially important for those who wear glasses.

(MTD 11)